Tong Chen

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Research Interests

Machine learning and AI for medical imaging, with a focus on Alzheimer's disease (AD) and Lewy body dementia (LBD). My research develops self-supervised representation learning methods for longitudinal brain MRI, models disease progression across the cognitive impairment continuum, and builds interpretable models on large multi-modal clinical cohorts.

Education

Ph.D. in Computer Science Aug 2023 – Present (expected Dec 2026)
The University of Texas at Arlington *Advisor: Dr. Dajiang Zhu*

M.S. in Computer Science Jan 2021 – Dec 2022
Kennesaw State University

B.S. in Computer Science Aug 2012 – May 2018
Kennesaw State University

Research Experience

Graduate Research Assistant — The University of Texas at Arlington Aug 2023 – Present
Advisor: Dr. Dajiang Zhu

- Developed graph neural network methods for multi-class dementia classification and continuous disease-progression modeling using large multi-cohort neuroimaging data.
- Built sMRI and DTI preprocessing and analysis pipelines supporting multi-cohort studies across the lab.
- Published first-author work at MICCAI and AAIC, and co-authored papers across MICCAI, ISBI, and related venues.
- Assisted in preparing NIH grant proposals.

Research Assistant — University of North Carolina at Chapel Hill May 2025 – Aug 2025
Advisor: Dr. Gang Li

- Developed cortical surface-based machine learning methods for brain MRI analysis using Human Connectome Project (HCP) data.

Research Assistant — University of Nevada, Las Vegas Mar 2023 – May 2023
Advisor: Dr. Junggab Son

- Researched interpretability methods for deep learning and generative models.

Graduate Research Assistant — Kennesaw State University Jan 2021 – Dec 2022
Advisors: Dr. Junggab Son; Dr. Jiho Noh

- Researched computer vision, natural language processing, generative models, and security of deep learning algorithms; published at ICMLA 2022.

Publications

(* = equal contribution.)

Conference Publications

In computer science and medical imaging, top conferences such as MICCAI and ISBI are fully peer-reviewed *archival publications.*

1. **Tong Chen**, Minheng Chen, Jing Zhang, Chao Cao, Li Su, Tianming Liu, and Dajiang Zhu. "HiLoGraph: Hierarchical-Longitudinal Brain Network Representation Learning." *Intl. Conf. on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2026. **(Early accept, top 9%)**
2. **Tong Chen**, Minheng Chen, Jing Zhang, Yan Zhuang, Chao Cao, Xiaowei Yu, Yanjun Lyu, Lu Zhang, Li Su, Tianming Liu, and Dajiang Zhu. "A Unified Continuous Staging Framework for Alzheimer's Disease and Lewy Body Dementia via Hierarchical Anatomical Features." *Intl. Conf. on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, Daejeon, South Korea, 2025. **(Early accept, top 9%)**
3. Chao Cao*, **Tong Chen***, Nan Zhao*, Minheng Chen, Minghao Qu, Zhiyuan Zhang, Xin Shi, Xiang Li, Tianming Liu, and Dajiang Zhu. "Gyral-Sulcal-Net: An Integrated Network Representation of Brain Folding Patterns." *IEEE Intl. Symposium on Biomedical Imaging (ISBI)*, 2026. **(Co-first author)**
4. Minheng Chen, **Tong Chen**, Chao Cao, Jing Zhang, Li Su, Tianming Liu, and Dajiang Zhu. "Folding-Fiducial Geodesic Representation for Morphometric Similarity Networks in Neurodegenerative Disease Characterization." *IEEE Intl. Symposium on Biomedical Imaging (ISBI)*, 2026.
5. Jing Zhang, Yanjun Lyu, Chao Cao, Minheng Chen, **Tong Chen**, Li Su, Tianming Liu, and Dajiang Zhu. "Integrating Population and Individual Brain Network with Mixture-of-Experts for Mild Cognitive Impairment Study." *IEEE Intl. Symposium on Biomedical Imaging (ISBI)*, 2026.
6. Minheng Chen, Xiaowei Yu, Jing Zhang, **Tong Chen**, Chao Cao, Yan Zhuang, Yanjun Lyu, Lu Zhang, Tianming Liu, and Dajiang Zhu. "Core-Periphery Principle Guided State Space Model for Functional Connectome Classification." *Intl. Conf. on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025.
7. Xiaowei Yu, Jing Zhang, **Tong Chen**, Yan Zhuang, Minheng Chen, Chao Cao, Yanjun Lyu, Lu Zhang, Li Su, Tianming Liu, and Dajiang Zhu. "Domain-Adaptive Diagnosis of Lewy Body Disease with Transferability Aware Transformer." *Intl. Conf. on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025.
8. Yan Zhuang, Minheng Chen, Chao Cao, **Tong Chen**, Jing Zhang, Xiaowei Yu, Yanjun Lyu, Lu Zhang, Tianming Liu, and Dajiang Zhu. "GyralNet Subnetwork Partitioning via Differentiable Spectral Modularity Optimization." *Intl. Workshop on Machine Learning in Medical Imaging (MLMI)*, pp. 477–486, 2025.
9. Jing Zhang, Xiaowei Yu, Minheng Chen, Lu Zhang, **Tong Chen**, Yan Zhuang, Chao Cao, Yanjun Lyu, Li Su, Tianming Liu, and Dajiang Zhu. "Bridging Brain Connectomes and Clinical Reports for Early Alzheimer's Disease Diagnosis." *Intl. Workshop on Machine Learning in Medical Imaging (MLMI)*, pp. 604–614, 2025.
10. Jing Zhang, Xiaowei Yu, **Tong Chen**, Chao Cao, Minheng Chen, Yan Zhuang, Yanjun Lyu, Lu Zhang, Li Su, Tianming Liu, and Dajiang Zhu. "BrainNet-MoE: Brain-Inspired Mixture-of-Experts Learning for Neurological Disease Identification." *IEEE Intl. Conf. on Data Mining Workshops (ICDMW)*, pp. 1995–2001, 2025.
11. Jing Zhang, Yanjun Lyu, Xiaowei Yu, Lu Zhang, Chao Cao, **Tong Chen**, Minheng Chen, Yan Zhuang, Tianming Liu, and Dajiang Zhu. "Classification of Mild Cognitive Impairment Based on Dynamic Functional Connectivity Using Spatio-Temporal Transformer." *IEEE Intl. Symposium on Biomedical Imaging (ISBI)*, 2025.
12. Jing Zhang, Xiaowei Yu, Yanjun Lyu, Lu Zhang, **Tong Chen**, Chao Cao, Yan Zhuang, Minheng Chen, Tianming Liu, and Dajiang Zhu. "Brain-Adapter: Enhancing Neurological Disorder Analysis with Adapter-Tuning Multimodal Large Language Models." *IEEE Intl. Symposium on*

Biomedical Imaging (ISBI), 2025.

13. Minheng Chen, Chao Cao, **Tong Chen**, Yan Zhuang, Jing Zhang, Yanjun Lyu, Xiaowei Yu, Lu Zhang, Tianming Liu, and Dajiang Zhu. "Using Structural Similarity and Kolmogorov-Arnold Networks for Anatomical Embedding of Cortical Folding Patterns." *IEEE Intl. Symposium on Biomedical Imaging (ISBI)*, 2025.
14. Yanjun Lyu, Jing Zhang, Lu Zhang, **Tong Chen**, Xiaowei Yu, Minheng Chen, Yan Zhuang, Chao Cao, Tianming Liu, and Dajiang Zhu. "CA-VQVAE: Cortical Folding Aware Numerical Representation of White-Matter Structure." *IEEE Intl. Symposium on Biomedical Imaging (ISBI)*, 2025.
15. Xiaowei Yu, Lu Zhang, Chao Cao, **Tong Chen**, Yanjun Lyu, Jing Zhang, Tianming Liu, and Dajiang Zhu. "Gyri vs. Sulci: Core-Periphery Organization in Functional Brain Networks." *Intl. Conf. on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2024.
16. Chao Cao, Xiaowei Yu, Lu Zhang, **Tong Chen**, Yanjun Lyu, Tianming Liu, and Dajiang Zhu. "Enhancing Group-Wise Consistency in 3-Hinge Gyrus Matching via Anatomical Embedding and Structural Connectivity Optimization." *IEEE Intl. Symposium on Biomedical Imaging (ISBI)*, Athens, Greece, 2024.
17. **Tong Chen**, Jiho Noh, Luke Cranfill, John Morris, and Junggab Son. "Improving Fashion Attribute Classification Accuracy with Limited Labeled Data Using Transfer Learning." *21st IEEE Intl. Conf. on Machine Learning and Applications (ICMLA)*, Nassau, The Bahamas, 2022. **(Oral presentation, acceptance rate 32.4%)**

Journal Articles

1. Lu Zhang, Junqi Qu, Haotian Ma, **Tong Chen**, Tianming Liu, and Dajiang Zhu. "Exploring Alzheimer's Disease: A Comprehensive Brain Connectome-Based Survey." *Psychoradiology*, kkad033, 2024.

Conference Abstracts

1. **Tong Chen**, Minheng Chen, Yan Zhuang, Jing Zhang, Lu Zhang, Tianming Liu, Andrew M. Blamire, John-Paul Taylor, Li Su, John T. O'Brien, and Dajiang Zhu. "Representing the Cognitive Impairment Continuum of Alzheimer's Disease and Lewy Body Dementia with a Novel Finer-Scale Cortical Representation via Disease Embedding Tree." Alzheimer's Association International Conference (AAIC); abstract published in *Alzheimer's & Dementia*, 21, e102179, 2025.
2. Yanjun Lyu, Xiaowei Yu, Minghao Qu, **Tong Chen**, Nan Zhao, Xiang Li, and Lu Zhang. "Gyral-Sulcal Net: A Novel Brain Network Representation for Mild Cognitive Impairment Classification." Alzheimer's Association International Conference (AAIC); abstract published in *Alzheimer's & Dementia*, 21, e102417, 2025.

Presentations

- "A Unified Continuous Staging Framework for Alzheimer's Disease and Lewy Body Dementia via Hierarchical Anatomical Features." Poster, MICCAI 2025, Daejeon, South Korea.
- "Improving Fashion Attribute Classification Accuracy with Limited Labeled Data Using Transfer Learning." Oral presentation, ICMLA 2022, Nassau, The Bahamas.

Honors & Awards

- John Stephen Schuchman Memorial Outstanding PhD Student Scholarship Award, Dept. of Computer Science & Engineering, UT Arlington, 2026

Technical Skills

Programming: Python, MATLAB, LaTeX
ML / DL Frameworks: PyTorch, PyTorch Geometric, scikit-learn
Neuroimaging Tools: FreeSurfer, FSL, SPM12, CAT12, Connectome Workbench, DSI Studio, MRtrix3, medInria
Models: Graph Neural Networks, Transformers, CNNs, State-Space Models (Mamba), Multimodal LLMs
Datasets: ADNI, NACC, AABC, HCP, BCP, in-house LBD cohorts

Academic Service

- Program Committee Reviewer, Intl. Conf. on PErvasive Technologies Related to Assistive Environments (PETRA), 2026.